

Type I and Type II Errors, Power, and One- vs Two-tailed Tests

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1. Overview

Key point: Every test decision can be wrong in two ways: rejecting a true H_0 (Type I) or failing to reject a false H_0 (Type II); the direction of H_1 decides whether the rejection region sits in one tail or two.

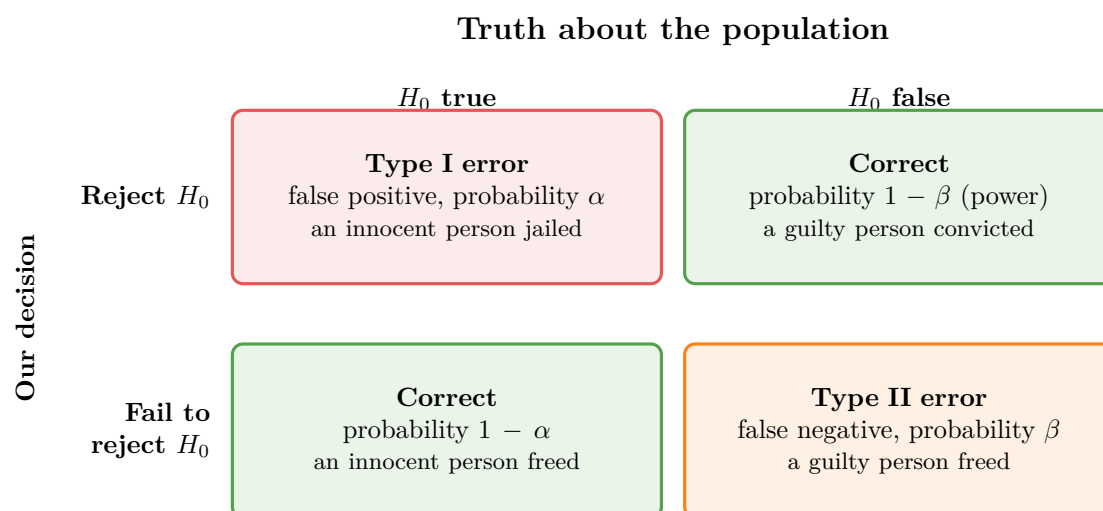


Figure 1: The four possible outcomes of a hypothesis test

A hypothesis test ends with one of two decisions, reject H_0 or fail to reject it (see the [null and alternative hypotheses Note](#)). The truth is also one of two: H_0 true or H_0 false. Figure 1 crosses them into four outcomes, two of which are errors.

This Note covers:

- Type I and Type II errors, with their probabilities α and β ;
- the power of a test, $1 - \beta$;
- the trade-off between the two errors;
- one-tailed and two-tailed tests, and the pros and cons of each;
- where hypothesis testing is used, in general and in machine learning.

2. Type I and Type II errors

Key point: A Type I error rejects a true H_0 (a false positive, probability α); a Type II error fails to reject a false H_0 (a false negative, probability β).

Whatever we decide, the decision may be wrong:

- we say “reject H_0 ”, but H_0 was actually true;
- we say “fail to reject H_0 ” for lack of evidence, but H_0 was actually false.

The other two outcomes are correct: rejecting a false H_0 , and failing to reject a true one.

2.1 Type I error

A **Type I error** occurs when the sample leads us to reject H_0 although it is in fact true. It is the mistake of finding a significant effect or relationship when there is none. It is also called a **false positive**.

In the courtroom of the [null and alternative hypotheses Note](#), H_0 says “no crime”. A Type I error convicts an innocent person.

The probability of a Type I error is the significance level α (see the [rejection region Note](#)). The researcher chooses α , so this risk is **under our control**. Lowering α from 5% to 1% pushes the critical values out and leaves more room for the innocent to go free.

2.2 Type II error

A **Type II error** occurs when the sample leads us to fail to reject H_0 although it is in fact false. The researcher fails to detect an effect or relationship that actually exists. It is also called a **false negative**.

In the courtroom: the accused did commit the crime, but the evidence was too weak and they walk free.

The probability of a Type II error is written β .

2.3 The error table

	H_0 true	H_0 false
Reject H_0	Type I error (probability α)	correct (probability $1 - \beta$)
Fail to reject H_0	correct (probability $1 - \alpha$)	Type II error (probability β)

Many books label the second row “accept H_0 ”. As the [null and alternative hypotheses Note](#) showed, the correct wording is “fail to reject H_0 ”: the test never proves H_0 .

The same two errors appear in classification. In the confusion matrix of the [accuracy and confusion matrix Note](#), a false positive is a Type I error and a false negative a Type II error. The link is direct: “positive” means “we flag an effect”, which in a test is “reject H_0 ”.

3. Power of a test

Key point: The power, $1 - \beta$, is the probability that the test rejects H_0 when H_0 is false: its ability to detect a real effect.

The **power of a test** is $1 - \beta$: the probability of correctly rejecting a false H_0 . A powerful test seldom misses a real effect. In practice, power matters more than β itself.

β and the power depend on how false H_0 is. Figure 2 works this out for the training program of the [rejection region Note](#) ($H_0 : \mu = 50$, $H_1 : \mu > 50$, $\sigma = 5$, $n = 30$).

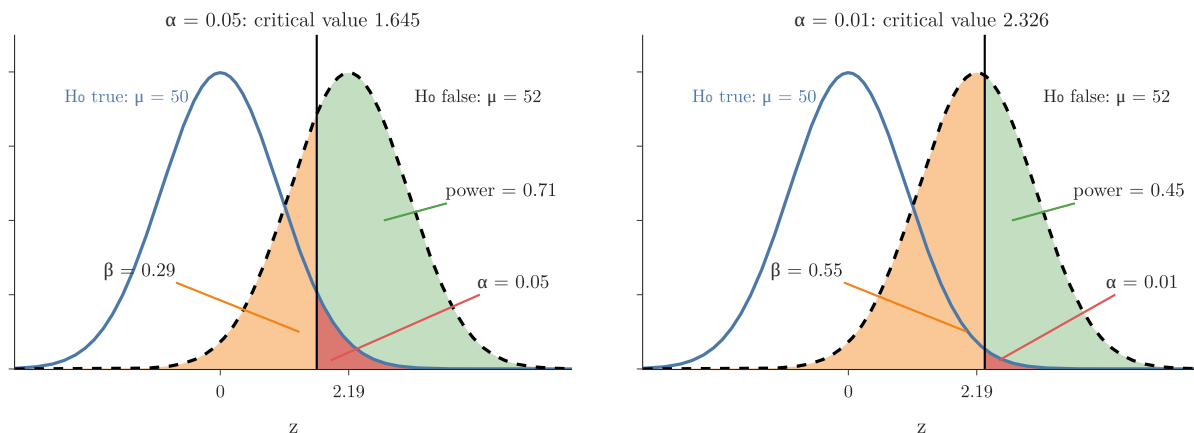


Figure 2: Type I error (red), Type II error (orange) and power (green) for the training-program test if the true mean is 52

Extra: Computing the power, assuming the true mean is 52.

1. **In words:** if $\mu = 52$, the z statistic is no longer centred at 0 but at the true difference in standard errors. Power is the chance that it still lands beyond the critical value.
2. **Formula:** with shift $d = (\mu_{\text{true}} - \mu_0)/(\sigma/\sqrt{n})$ and right-tailed critical value z_α ,

$$\beta = \Phi(z_\alpha - d), \quad \text{power} = 1 - \beta$$

where Φ is the standard normal CDF.

3. **Example:** $d = (52 - 50)/(5/\sqrt{30}) = 2.19$. At $\alpha = 0.05$:

$$\beta = \Phi(1.645 - 2.19) = \Phi(-0.55) = 0.29, \quad \text{power} = 0.71$$

So even if the training really added 2 cars a day, this test would miss it 29% of the time.

4. The trade-off between Type I and Type II errors

Key point: For a fixed sample, lowering α shrinks the rejection region and so raises β : we cannot push both errors down at once.

Lowering α protects against Type I errors, but it has a price. A smaller α widens the region where we fail to reject H_0 . A future test statistic is then more likely to land there, including when H_0 is false: more guilty people walk free.

Figure 2 shows it with numbers. For the same true mean of 52:

α	Critical value	β	Power
0.05	1.645	0.29	0.71

α	Critical value	β	Power
0.01	2.326	0.55	0.45

Cutting α from 5% to 1% almost doubles the chance of missing the effect. The two errors move in opposite directions, so we strike a balance. That is why 0.05 is the usual choice, unless a domain expert has a specific reason to change it.

Extra: The way to lower both errors is more data. Keeping $\alpha = 0.05$ and the true mean at 52, the power of the training test rises from 0.71 with 30 employees to 0.93 with 60 and 0.99 with 100. A larger sample shrinks the standard error, so the two curves of Figure 2 move apart.

5. One-tailed and two-tailed tests

Key point: H_1 with $>$ or $<$ gives a one-tailed test with all of α in one tail; H_1 with $\neq$ gives a two-tailed test with $\alpha/2$ in each tail.

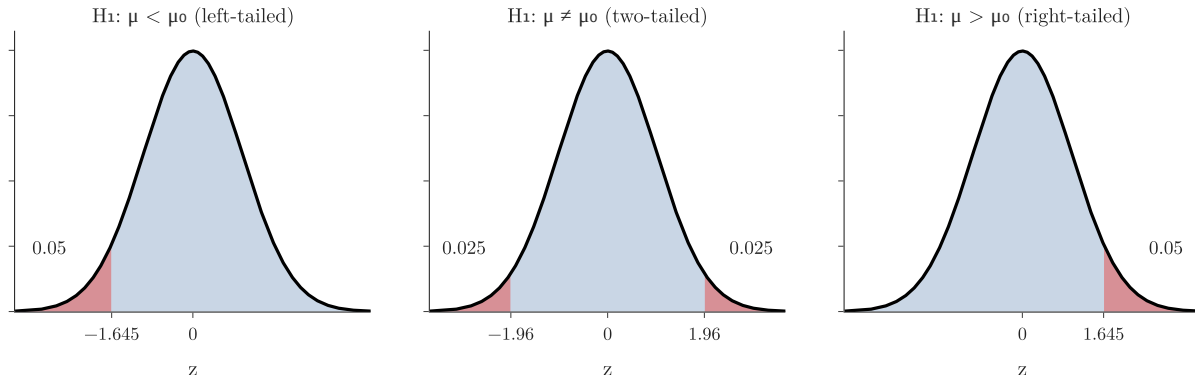


Figure 3: Rejection regions (red) at $\alpha = 0.05$ for the three kinds of alternative hypothesis

The form of H_1 fixes where the rejection region goes (Figure 3).

- A **one-tailed test** (one-sided test) is used when we test for an effect in one specific direction. H_1 contains an inequality sign:
 - **right-tailed**, $H_1: \mu > \mu_0$: “the new filming style increases the mean view duration”; “a new medication increases the average recovery rate compared with the existing one”;
 - **left-tailed**, $H_1: \mu < \mu_0$: “the new style decreases the mean view duration”.
- A **two-tailed test** (two-sided test) is used when we test for an effect in either direction. H_1 contains $\neq$: “the chips packets do not weigh 100 g”, more or less.

The rule of thumb: an H_1 that says “increases” or “decreases” is one-tailed; an H_1 that says “changes” or “differs” is two-tailed. At $\alpha = 0.05$ the critical values are 1.645 (or -1.645) for one tail and ± 1.96 for two (see the [rejection region Note](#)).

5.1 Two-tailed tests: pros and cons

Advantages:

- **They detect effects in both directions.** Wherever z falls, far left or far right, the test can see it. They suit situations where the direction is uncertain, or where we want to know about any difference between groups.
- **They are more conservative in each direction.** Each tail gets only $\alpha/2$, so a result in a given direction must be more extreme (beyond 1.96 instead of 1.645) to count.

Disadvantages:

- **They are less powerful** for an effect in a known direction. Because α is split, a larger effect is needed to reject H_0 . For the training program with a true mean of 52, a two-tailed test at 5% has power 0.59 against 0.71 for the right-tailed test.
- **They are not designed for a directional hypothesis.** The question “is the weight different from 100 g?” is not “is it lower?”.

Extra: Two corrections to common claims. First, a two-tailed test does not lower the overall Type I error rate: it is α in total, exactly as for a one-tailed test, only split across two tails. Second, when a two-tailed test rejects H_0 , the sign of z does tell the direction: $z = -2.5$ means the mean is below 100 g. What the two-tailed test gives up is power in each single direction, not the ability to report one.

5.2 One-tailed tests: pros and cons

Advantages:

- **They are more powerful in the chosen direction.** All of α sits in one tail, so the rejection region on that side is larger and a real effect there is easier to detect.
- **They fit a directional hypothesis.** When there is a strong theoretical or practical reason to expect one direction, such as a training program that can only help, the one-tailed test asks exactly that question.

Disadvantages:

- **They miss effects in the other direction.** If the new filming style actually lowers the view duration, a right-tailed test cannot see it: z lands far in the left tail, which is not a rejection region, and we “fail to reject” a null that is wrong in the opposite way.
- **They are easy to misuse.** The direction must be fixed before seeing the data.

Extra: The real Type I risk of one-tailed tests comes from choosing the tail after looking. With a correct one-tailed test the Type I error rate is exactly α . But if we look at z first and then test in the direction it points to, we reject whenever $|z| > 1.645$: in a simulation of 100,000 tests with a true H_0 , that happens 10.0% of the time, double the promised 5%.

6. Where hypothesis testing is used

Key point: Testing interventions, comparing groups, finding relationships, checking distributions, testing independence and A/B testing; in ML, comparing models, selecting features, tuning and checking assumptions.

6.1 In general

1. **Testing the effect of an intervention or treatment.** Does a new filming style raise view time? Does a new drug work? With σ known we use a z-test, otherwise a t-test (see the [one-sample t-test Note](#)).
2. **Comparing means and proportions** between two or more groups: customer satisfaction scores, conversion rates (proportions, i.e. percentages), employee performance, the average marks of section A and section B. The tools are t-tests (see the [two-sample t-tests Note](#)) and ANOVA.
3. **Relationships between variables.** Is the correlation between two numerical columns real or chance? Pearson's correlation coefficient (see the [covariance and correlation Note](#)) and Spearman's rank correlation each come with a test.
4. **Goodness of fit.** Does a dataset follow a normal, binomial or Poisson distribution? The chi-square test is one tool.
5. **Independence of categorical variables.** On the Titanic, are sex (male, female) and survival (1, 0) related, or independent? Is the type of product related to whether customers return it? The chi-square test answers this.
6. **A/B testing.** Two versions of a web page, thumbnail or product (A and B) are shown to random groups of users, and a test decides which converts or engages better (see the [machine learning development life cycle Note](#)). Tech companies run such tests constantly; it is hypothesis testing standardized for marketing, product development and website design.

6.2 In machine learning

1. **Model comparison.** We train XGBoost, a random forest and linear regression on the same data. Is one really better, or did it win by chance on this split? A paired t-test on the scores of the same cross-validation folds (see the [pipelines Note](#) for cross-validation) answers it; the [two-sample t-tests Note](#) runs one.
2. **Feature selection.** Which features are related to the target, and which can we drop (see the [what is feature engineering Note](#))? A t-test, chi-square test or ANOVA between each feature and the target gives a p-value per feature. scikit-learn's `SelectKBest` with `f_classif` (ANOVA) or `chi2` does exactly this.
3. **Hyperparameter tuning.** Two settings of a model give different cross-validation scores (see the [random forest tuning Note](#)). A test says whether one setting is significantly better.
4. **Checking model assumptions.** Linear regression assumes, among other things, normally distributed residuals (see the [linear regression assumptions Note](#)). The Shapiro-Wilk test checks this; if the assumptions fail, the model may not suit the data.

Libraries such as scikit-learn usually run these tests for us. Knowing what happens behind the scenes lets us use those tools better and make better-educated decisions.

7. Summary

	One-tailed	Two-tailed
H_1	$\mu > \mu_0$ or $\mu < \mu_0$	$\mu \neq \mu_0$
Rejection region	all of α in one tail	$\alpha/2$ in each tail
Critical value, $\alpha = 0.05$	1.645 or -1.645	± 1.96
Strength	more power in the chosen direction	detects effects in both directions
Weakness	blind to the other direction	less power in each direction

- Type I error: reject a true H_0 (false positive), probability α , chosen by us.
- Type II error: fail to reject a false H_0 (false negative), probability β .
- Power = $1 - \beta$: the chance of detecting a real effect; 0.71 for the training test if $\mu = 52$.
- Lowering α raises β ; only more data lowers both.
- Fix the direction of H_1 before looking at the data.

8. Key terms

Term	Meaning
Type I error (false positive)	Rejecting H_0 when it is actually true; its probability is α
Type II error (false negative)	Failing to reject H_0 when it is actually false; its probability is β
β	The probability of a Type II error
Power of a test	$1 - \beta$: the probability of rejecting H_0 when it is false
One-tailed test (one-sided test)	A test whose H_1 has a direction ($>$ or $<$), with the whole rejection region in one tail
Right-tailed and left-tailed test	One-tailed tests for $H_1 : \mu > \mu_0$ and $H_1 : \mu < \mu_0$
Two-tailed test (two-sided test)	A test whose H_1 is $\neq$, with $\alpha/2$ in each tail